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Machine Translation Project – Ndyuka & Backtranslation

By Group 5 – Yifan Hu, Steinar Grässel, Daan Brugmans



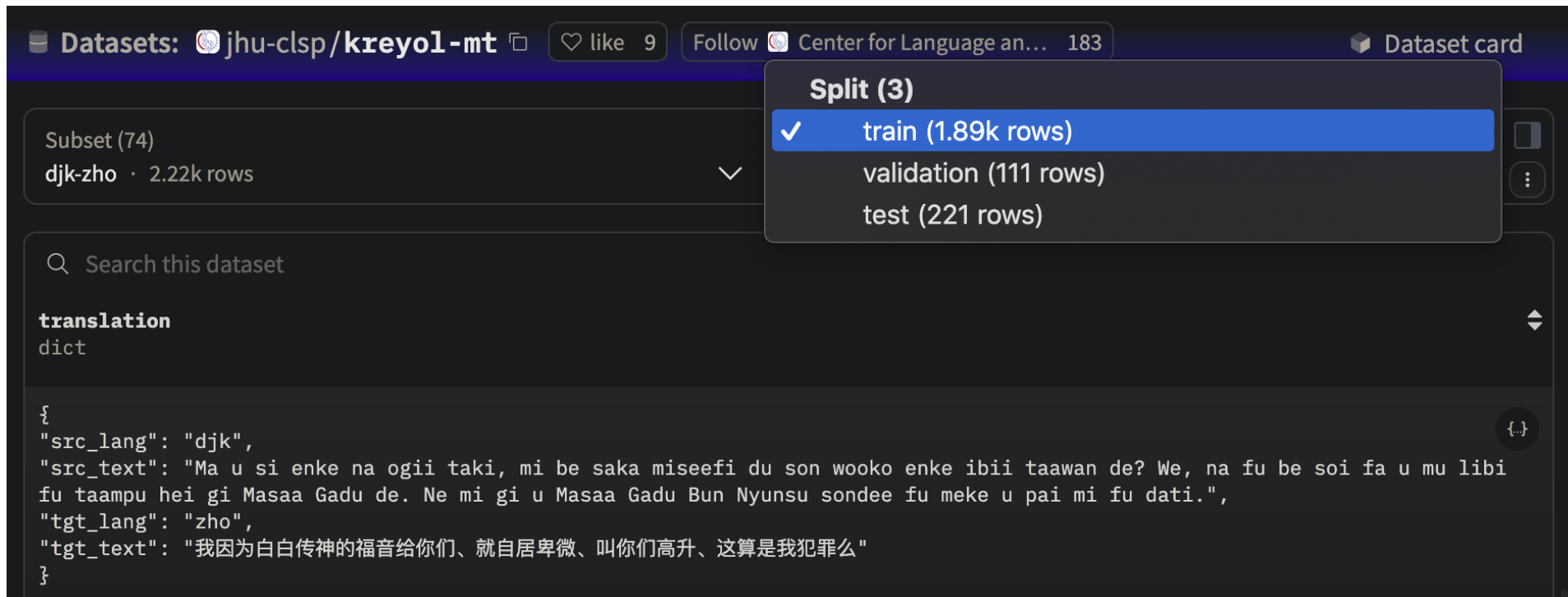
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Project Outline

- Machine Translation for Ndyuka
 - Creole in Suriname + French Guyana
- From Ndyuka to Chinese & German
- Research focus: backtranslation
 - Translate from target to source to gain more sentence pairs
 - High-resource vs. Low-resource
- Research question: *What is the change in performance for our Ndyuka to Chinese/German machine translation systems when backtranslated sentence pairs are added to the training data?*
- Three models/pipelines are considered:
 1. SMT for Ndyuka → Chinese
 2. SMT for Ndyuka → German
 3. NMT for Ndyuka → Chinese + German



SMT Ndyuka → Chinese: Dataset



Datasets:  [jhu-clsp/kreyol-mt](#)   like 9  Follow  Center for Language and Cognition 183  Dataset card

Subset (74)
djk-zho · 2.22k rows

Split (3)

- ✓ train (1.89k rows)
- validation (111 rows)
- test (221 rows)

Search this dataset

translation
dict

```
{
  "src_lang": "djk",
  "src_text": "Ma u si enke na ogii taki, mi be saka miseefi du son wooko enke ibii taawan de? We, na fu be soi fa u mu libi fu taampu hei gi Masaa Gadu de. Ne mi gi u Masaa Gadu Bun Nyunsu sondee fu meke u pai mi fu dati.",
  "tgt_lang": "zho",
  "tgt_text": "我因为白白传神的福音给你们、就自居卑微、叫你们高升、这算是我犯罪么"
}
```



SMT Ndyuka → Chinese: Dataset (More examples)

- Weird archaic sentences (either poor translation or religious wording?)

1 我因为白白传神的福音给你们、就自居卑微、叫你们高升、这算是我犯罪么
2 你们这瞎眼领路的、蠅虫你们就滤出来、骆驼你们倒吞下去
3 因为女人原是由男人而出、男人也由女人而出、但万有都是出乎神
4 亚基帕对保罗说、准你为自己辩明。于是保罗伸手分诉说
5 污秽的言语、一句不可出口、只要随事说造就人的好话、叫听见的人得益处
6 用和平彼此联络、竭力保守圣灵所赐合而为一的心
7 那暗昧无益的事、不要与人同行、倒要责备行这事的人
8 因为主的道从你们那里已经传扬出来、你们向神的信心不但在马其顿和亚该亚、就是在各处、也都传开了。所以不用我们说甚么话
9 然而我今日成了何等人、是蒙神的恩才成的。并且他所赐我的恩、不是徒然的。我比众使徒格外劳苦。这原不是我、乃是神的恩与我同在
10 这些原是后事的影儿。那形体却是基督
11 因此、你们是大有喜乐、但如今、在百般的试炼中暂时忧愁
12 在这地方神并没有给他产业、连立足之地也没有给他。但应许要将这地赐给他和他的后裔为业。那时他还没有儿子
13 你要提醒众人、叫他们顺服作官的、掌权的、遵他的命、预备行各样的善事
14 我先前心里难过痛苦、多多的流泪、写信给你们。不是叫你们忧愁、乃是叫你们知道我格外的疼爱你们
15 内中有抹大拉的马利亚、又有雅各和约西的母亲马利亚、并有西庇太两个儿子的母亲
16 我为这福音奉派作传道的、作使徒、作师傅
17 亲爱的弟兄阿、你们要记念我们主耶稣基督之使徒从前所说的话
18 无论何人、不要求自己的益处、乃要求别人的益处
19 你们中间的争战斗殴、是从那里来的呢。不是从你们百体中战斗之私欲来的么
20 忽然有主的一个使者、站在旁边、屋里有光照耀。天使拍彼得的肋旁、拍醒了他、说、快快起来。那铁链就从他手上脱落下来



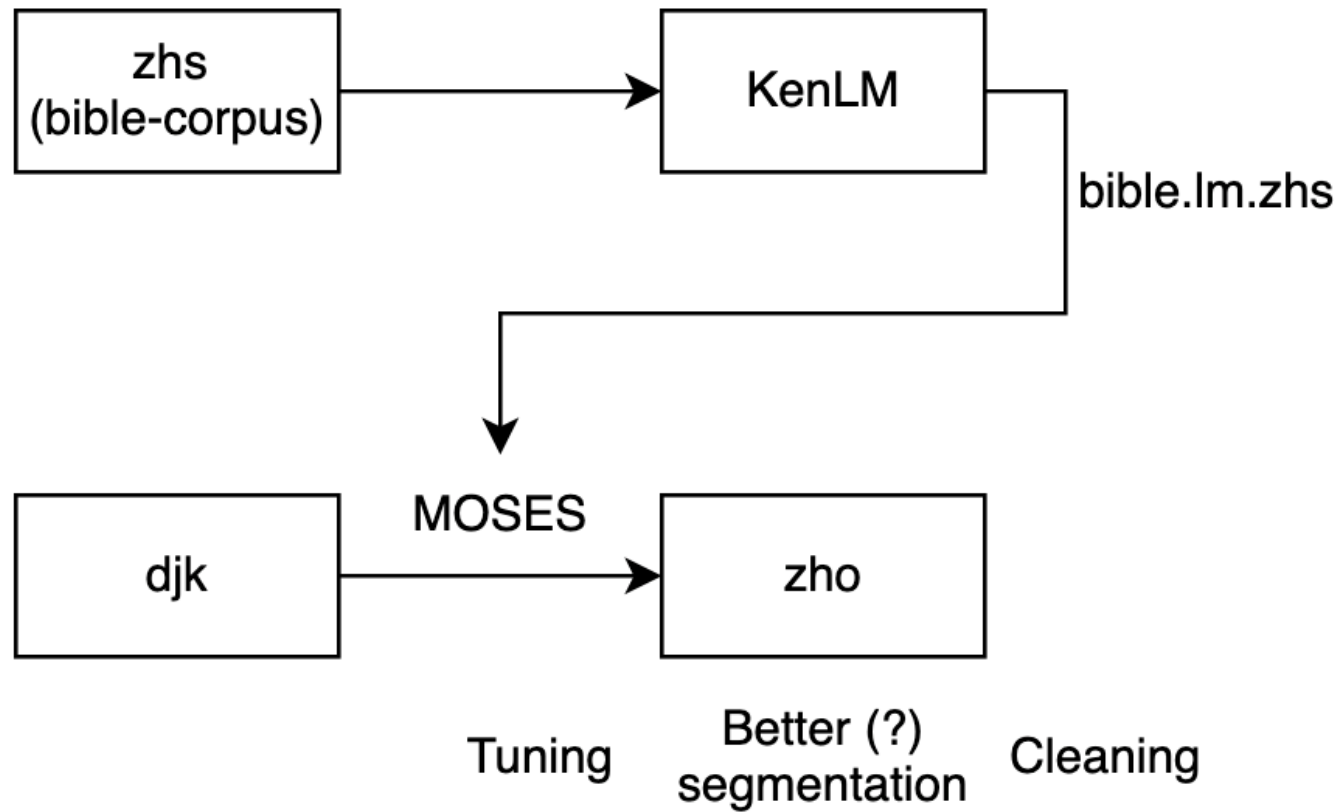
SMT Ndyuka → Chinese: Extra data

- <https://github.com/christos-c/bible-corpus>
- xml format in **Traditional Chinese**
- Used from n-gram LM model & extra data for backtranslation
- But why?
 - It is relatively **large** -> 677919 words count
 - It is **from Bible** -> The original text is also from Bible
 - It is **already segmented!!!**

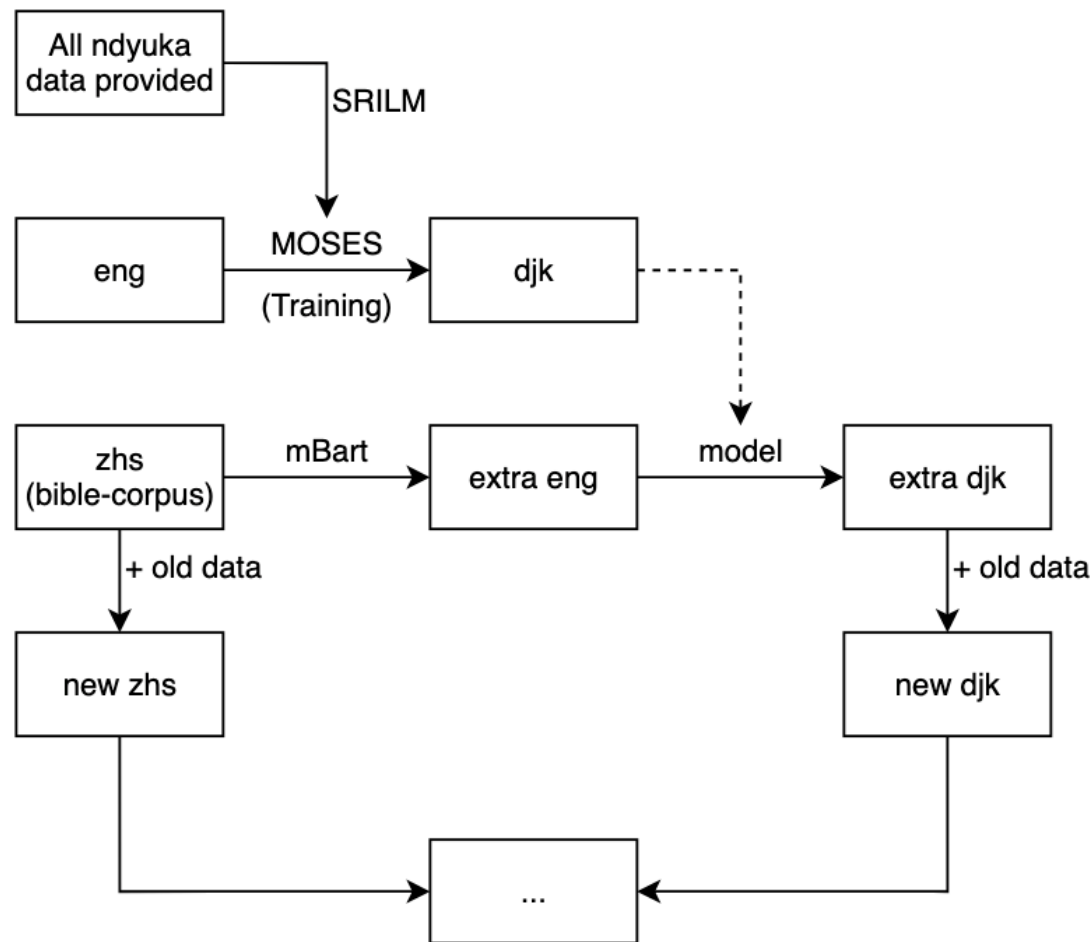
```
<body id="Bible" lang="zh">
  <div id="b.GEN" type="book">
    <div id="b.GEN.1" type="chapter">
      <seg id="b.GEN.1.1" type="verse">
        起初 神 創造 天地
      </seg>
      <seg id="b.GEN.1.2" type="verse">
        地是 空虛 混沌 . 淵面 黑暗 . 神 的 靈運 行在 水面上
      </seg>
      <seg id="b.GEN.1.3" type="verse">
        神說、要有光、就有了光
```



SMT Ndyuka → Chinese: Pipeline



SMT Ndyuka → Chinese: Pipeline



- This is the pipeline for creating data for backtranslation
- The new Chinese and Ndyuka data will then go through the same process as the previous one



SMT Ndyuka → Chinese: Result

- Baseline: cleaned + tuning (BLEU 2.17)
- Cleaning itself might not so helpful in this case (1.98 vs. 1.93)
- Segmentation does matter

```
$ multi-bleu.perl ../../corpora/zho-djk/might_better_seg/test.tk.zho < test.out.zho
BLEU = 0.89, 13.4/2.0/0.4/0.1 (BP=1.000, ratio=2.545, hyp_len=10766, ref_len=4231)
```
- Back translation (cleaned + tuning): BLEU 2.45



SMT Ndyuka → Chinese: Does it work (for Chinese)?

- Maybe
- Improvement about 0.3, but with 31k of extra sentence, taking the whole weekend (8h for zhs to eng on UPPMAX with GPU, plus some other hours to train and test
- Am I good at doing translation?
- Will the quality of original data affect?
(as we still need segmentation for traditional method)
- If segmentation matters, can we get rid of it? -> alignment issue
- Is BLEU score reliable?
- Is the processing correct?
 - Same data, same commands, eng to djv scored only 0.61, but **djv to eng got above 10** ???
- Garbage in, garbage out

Slurm Job_id=9980596 Name=mt_proj Ended, Run time 08:21:09, COMPLETED, ExitCode 0

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SMT Ndyuka → German

- Pipeline essentially analogous to Ndyuka → Chinese
- Baseline generated with a ready-to-use German LM (from CMU Sphinx)
 - BLEU: 1.19
- Backtranslation uses Nduyka LM trained with ~12k sentences from the Ndyuka ↔ English data in the Kreyol-MT dataset
 - Assumption: This pairing is the biggest one for Ndyuka, and will encompass most sentences within the smaller other pairings
 - Translate 1640 (size of original training data) randomly chosen verses from ~31k bible verses in German bible corpus
- Enhancing training data with 1640 additional training pairs leads to BLEU: 1.35

djk-ara (1.16k rows)
djk-ceb (2.04k rows)
djk-deu (1.86k rows)
djk-eng (11.8k rows)
djk-fra (7.69k rows)
djk-nep (1.9k rows)
djk-zho (2.22k rows)



SMT Ndyuka → German

- Lean into domain-specificity: Train a German “*Bible-LM*” using the ~31k verses
 - BLEU with Bible-LM: 3.00
 - BLEU with Bible-LM & additional data from backtranslation: 3.49

Pending ideas & thoughts

- Check if using the (tiny) validation set makes a difference
- Supplement Bible-LM with out-of-domain data (~31k lines is not a lot)
- Linguistic analysis
 - Nduyuka exhibits reduplication:
 - *wanten wanten* → *alsobald alsobald*
 - *sete sete* → *dieser ist dieser*



NMT Ndyuka → Chinese + German

- NMT from the ground up infeasible
- However, authors of creole dataset also published model
 - Multilingual neural model
- Baseline metrics:
 - German: 16.2 BLEU, 43 ChrF
 - Chinese: not yet obtained
- Continue finetuning of NN on dataset of backtranslations
 - PEFT/LoRA?
- Use same resources for backtranslating as SMT models
- Results are to follow





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